MODULE 1 · LECTURE 28

# AI in Sorting and Food Processing Applications

*How machine vision and AI grade fruits, vegetables and grains by size, colour, defects and internal quality; how an in-line sorter and an optical colour sorter work; how grading accuracy is measured with a confusion matrix; and where AI supports food processing — with worked examples, a grading program and India's AGMARK standards.*

### Learning objectives

After this lecture you should be able to:

1. Explain why agricultural produce is sorted and graded, and the role of AGMARK in India.
2. Describe the parts of an in-line machine-vision grading line and the features it measures.
3. Explain how deep learning and near-infrared sensing assess external and internal quality.
4. Explain the working of an optical colour sorter for grain and calculate its mass balance.
5. Evaluate a grading system with a multi-class confusion matrix.
6. Describe applications of AI across food processing and their limitations.

Scope: Lecture 25 used computer vision to find diseases and weeds in the field, and Lecture 27 to harvest fruit. This lecture follows the produce into the pack-house, mill and factory, where AI sorts, grades and inspects it. Digital initiatives and India's data ecosystem for agriculture follow in Lectures 30 and 31.

## 1. Why sort and grade?

- **Price**: buyers pay according to quality; uniform lots of known grade sell better and travel further.
- **Standards and trade**: grades describe quality in agreed terms, so buyers can trade without seeing every lot.
- **Loss reduction**: removing damaged or rotting items early stops spoilage spreading in storage and transport.
- **Processing**: factories need raw material of uniform size and ripeness; undersized but sound produce goes to processing instead of waste.
- **Food safety**: removing mouldy, insect-damaged or contaminated items and foreign matter protects consumers.

In India, the **AGMARK** quality certification mark acts as a third-party guarantee of certified quality, with legal backing from the Agricultural Produce (Grading and Marking) Act, 1937, as amended in 1986. The Directorate of Marketing and Inspection (DMI) provides grading and certification for export and domestic trade, with grade standards covering 248 commodities, and issues guidelines for quality-control laboratories at e-NAM mandis (Directorate of Marketing & Inspection [DMI], n.d.).

## 2. Machine-vision sorting and grading

Manual grading is slow, tiring and subjective. Artificial vision systems inspect fruits and vegetables automatically for grading, for estimating quality from external or internal features, and for monitoring produce in storage. Using ultraviolet or near-infrared light they can see defects the human eye cannot; hyperspectral systems detect components or damage visible only at particular wavelengths; and in-line systems inspect huge numbers of items individually while recording statistics for the whole batch (Cubero et al., 2011).

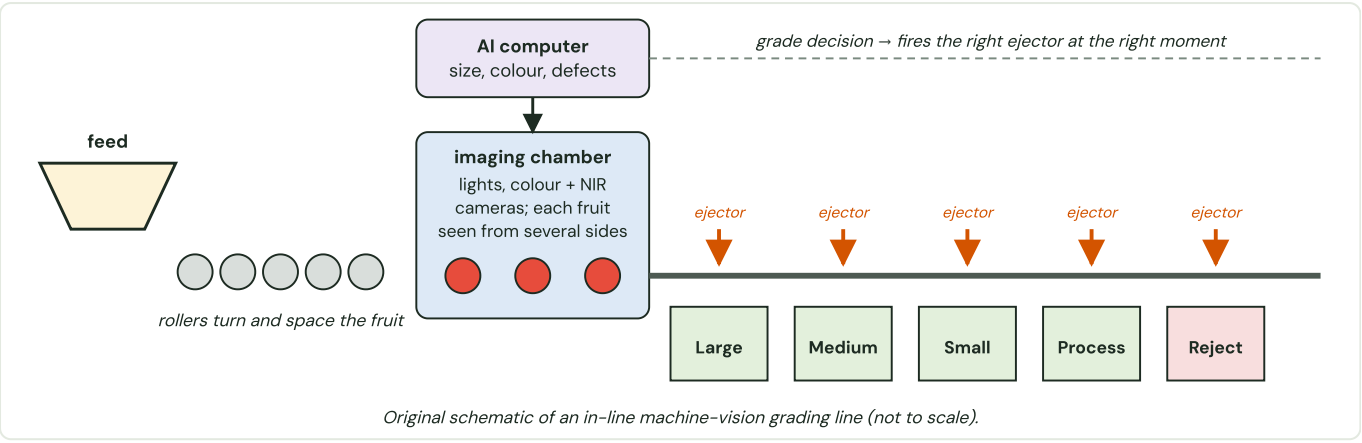



*Figure 1. An in-line machine-vision grading line. Original schematic.*

| Stage | What happens |
|---|---|
| Feeding and singulation | Produce is spread into lanes, one item per cup or roller, so each can be seen and tracked |
| Imaging | Uniform lighting; colour and near-infrared cameras capture each item, often from several sides as rollers turn it |
| Analysis | Software measures size, shape and colour and detects defects; AI models classify the item |
| Decision | Grade rules or a trained model assign the grade |
| Ejection | When the item reaches the right outlet, a cup tilts or an air jet or finger pushes it off; timing comes from conveyor speed |

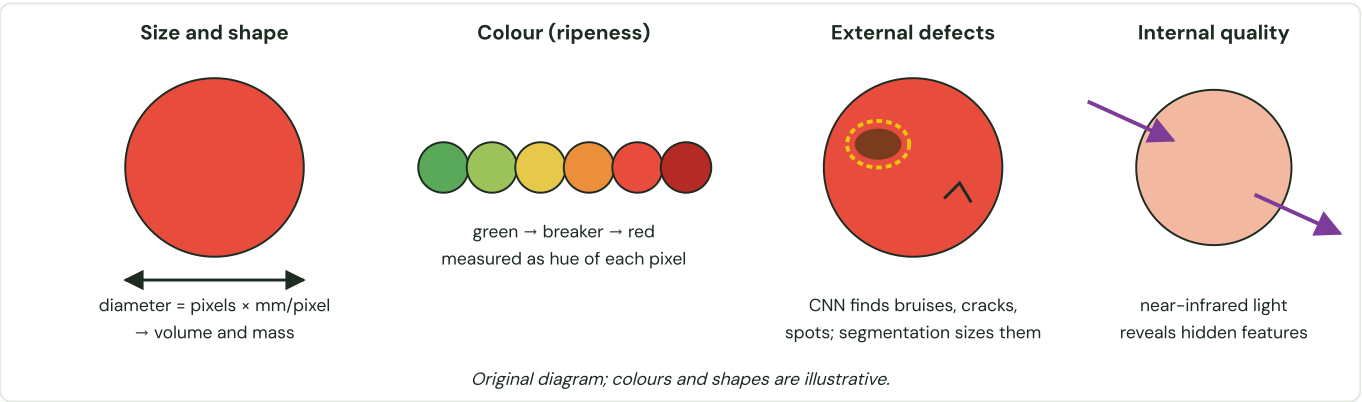



*Figure 2. Features measured by a vision grading system. Original diagram.*

Machine learning and deep learning models identify the type and quality of food from images, so machine-vision systems can grade food, locate defective spots or foreign objects, and remove impurities (Zhu et al., 2021). As in the field (Lecture 25), CNNs trained on many labelled images recognise defects that simple colour rules miss.

**Worked example 1 — from pixels to grams**
The camera sees 0.2 mm per pixel. A tomato 310 pixels across has diameter 310 × 0.2 = **62 mm**.
Treating it as a sphere of density 1.0 g/cm³: mass = 1.0 × (π ÷ 6) × 6.2³ = **125 g**.
If the diameter is over-read by 1 mm (63 mm), the estimate becomes 131 g — 4.9% too high. Because mass depends on the cube of size, a 1.6% error in diameter becomes about a 5% error in mass: precise size measurement matters.

> **Worked example 2 — capacity of a grading line**
> Cups 0.1 m apart on a conveyor moving at 1.0 m/s carry 1.0 ÷ 0.1 = 10 fruits per second per lane = 36,000 per hour.
> A 4-lane line handles 144,000 fruits per hour; at 125 g each that is **18 t/h**.
> An outlet 1.2 m after the camera is reached 1.2 s after imaging: the computer must finish its decision and the controller must fire the ejector exactly then.

*Sizes, densities, speeds and spacing in worked examples 1–3 are illustrative assumptions.*

## 3. Worked example: grading tomatoes

The program simulates 2,000 tomatoes of varying size, 8% of them with defects. A grading rule (an example, not an official standard) sends defective fruit to Reject and grades sound fruit by diameter. The vision system measures diameter with an error and finds defects with a CNN that catches 92% of them and wrongly flags 2% of sound fruit. The program compares the machine's grades with the true grades.

### Machine-vision grading of tomatoes with a multi-class confusion matrix in Python (Lecture 8)

```
import math, random

random.seed(21)
GRADES = ["Large", "Medium", "Small", "Process", "Reject"]

def grade(diameter_mm, defect):
    """Example grading rule: defects are rejected; sound fruit graded by diameter."""
    if defect:
        return "Reject"
    if diameter_mm >= 67:
        return "Large"
    if diameter_mm >= 57:
        return "Medium"
    if diameter_mm >= 47:
        return "Small"
    return "Process"                    # too small for the fresh market

# 1. A batch of 2,000 tomatoes: true size and true defects (bruises, cracks, spots)
fruits = [(random.gauss(60, 8), random.random() < 0.08) for _ in range(2000)]

# 2. The vision system: diameter measured from the image with some error,
#    defects found by a CNN that catches 92% of them and wrongly flags 2% of sound fruit
def inspect(d, defect, size_error_mm):
    d_seen = d + random.gauss(0, size_error_mm)
    flagged = random.random() < (0.92 if defect else 0.02)
    return d_seen, flagged

for err in (0.5, 1.0, 2.0):
    table = {(a, b): 0 for a in GRADES for b in GRADES}
    for d, defect in fruits:
        table[(grade(d, defect), grade(*inspect(d, defect, err)))] += 1
    correct = sum(table[(g, g)] for g in GRADES)
    print(f"size error {err} mm: {100 * correct / len(fruits):.1f}% graded correctly")
    if err == 1.0:
        print("   rows: true grade; columns: grade given by the machine")
        print("                  " + " ".join(f"{g:>7s}" for g in GRADES))
        for a in GRADES:
            print(f"   {a:14s} " + " ".join(f"{table[(a, b)]:7d}" for b in GRADES))

d = 62
mass = 1.0 * math.pi / 6 * (d / 10) ** 3       # sphere, density 1.0 g/cm3
print(f"Estimated mass of a {d} mm tomato: {mass:.0f} g")

Output:
size error 0.5 mm: 93.3% graded correctly
size error 1.0 mm: 90.2% graded correctly
   rows: true grade; columns: grade given by the machine
                   Large  Medium   Small Process  Reject
   Large             301      16       0       0       4
   Medium             29     743      42       0      18
   Small               0      34     514      11      11
   Process             0       0       8     104       6
   Reject              2       9       5       0     143
size error 2.0 mm: 84.7% graded correctly
Estimated mass of a 62 mm tomato: 125 g
```

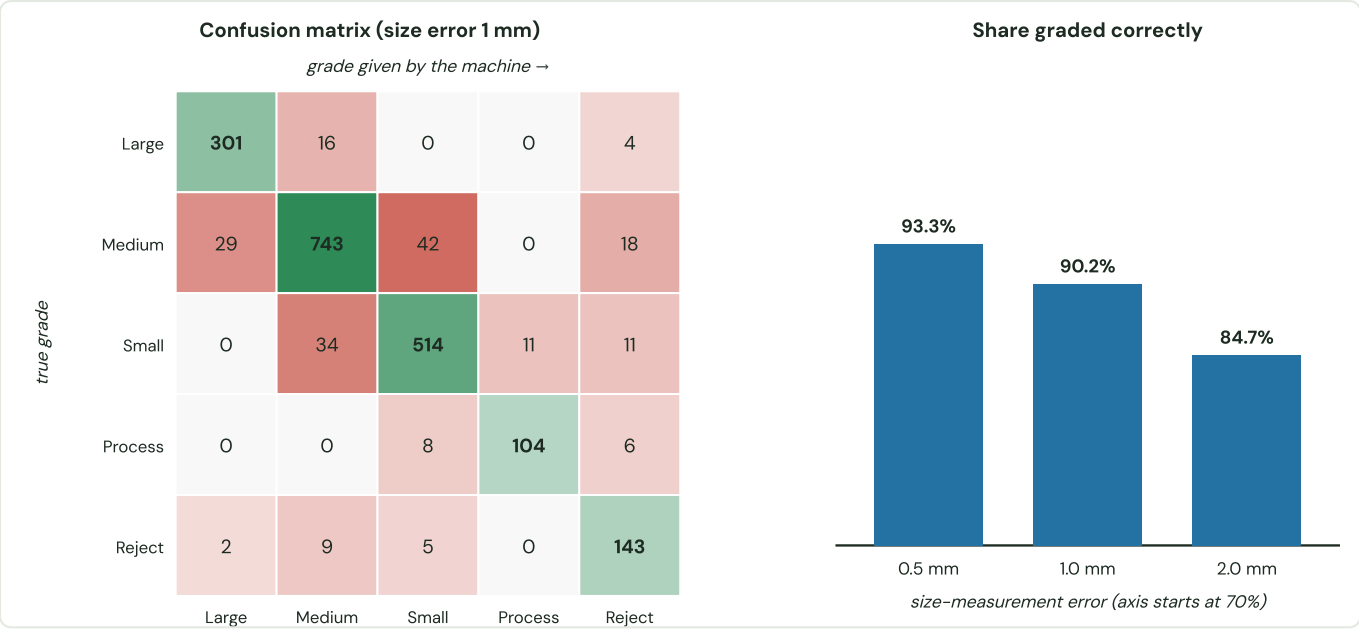



*Figure 3. Left: confusion matrix of true and machine grades. Right: share graded correctly for three sizes of measurement error. Original chart from the program's output.*

- A **multi-class confusion matrix** extends the 2 × 2 matrix of Lecture 24: the diagonal holds correct grades (1,805 of 2,000 = 90.2%); off-diagonal cells show which mistakes occur.
- Most size errors fall between **neighbouring** grades — 29 Medium fruits went to Large and 42 to Small — because fruits near a size limit are pushed across it by a small measurement error.
- 16 of the 159 defective fruits (10%) slipped into fresh-market grades, and 39 of the 1,841 sound fruits (2%) were wrongly rejected: defect detection, not size, causes the costliest errors.
- Halving the size error from 1.0 to 0.5 mm raised correct grading from 90.2% to 93.3%; doubling it to 2.0 mm lowered it to 84.7% — camera resolution and calibration matter.

*The program was run and the output shown is its actual output. The batch, grade limits and error rates are illustrative.*

## 4. Optical colour sorters for grain

Rice, pulses and other grains and seeds are sorted one grain at a time by **optical (colour) sorters**. Grains slide down a chute in a thin stream and fall past cameras; when a grain's colour or reflectance is wrong, an air jet blows it into the reject outlet. Surface features such as discoloration caused by fungi are generally detectable in the visible range, whereas internal attributes such as insect infestation are detectable in the near-infrared; the spectral differences are often subtle, so choosing the right wavelengths and sorting criterion is important (Pasikatan & Dowell, 2001).

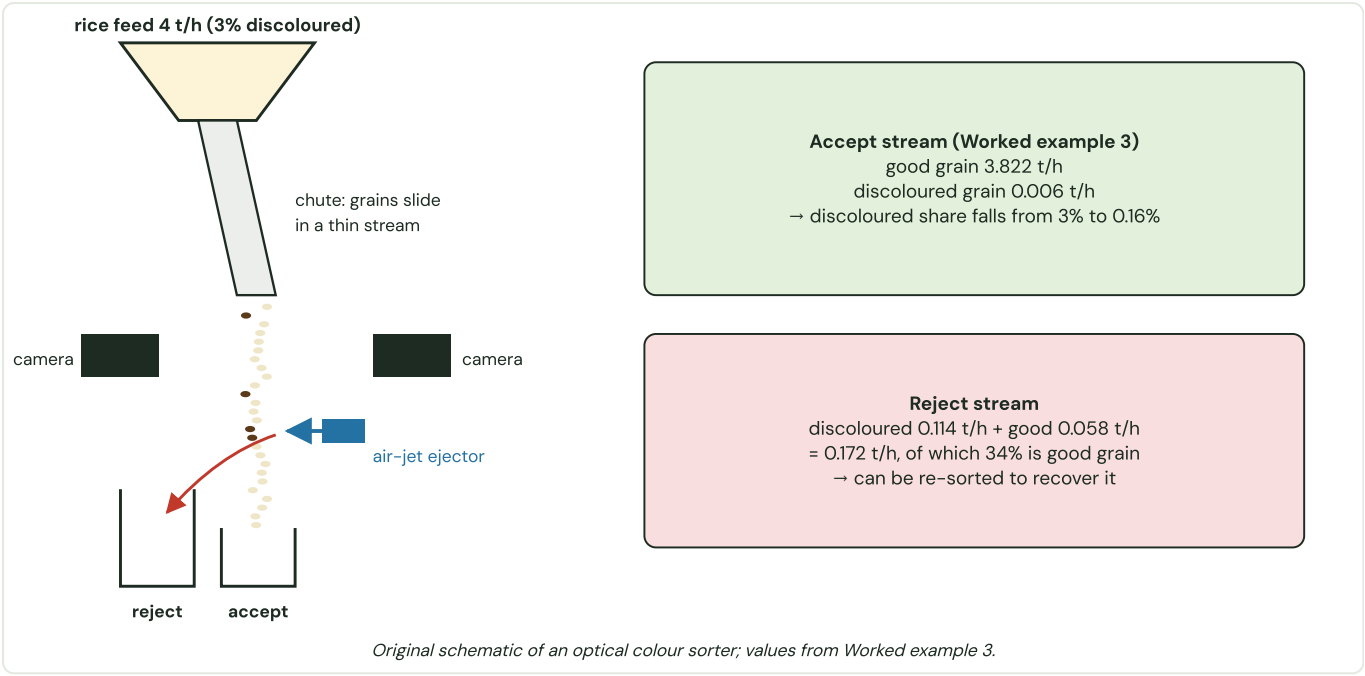



*Figure 4. An optical colour sorter and the streams it produces. Original schematic.*

**Worked example 3 — mass balance of a rice colour sorter**
Feed 4 t/h with 3% discoloured grain. The sorter rejects 95% of discoloured grain and, by mistake, 1.5% of good grain.
Accept stream: good grain 4 × 0.97 × 0.985 = 3.822 t/h; discoloured grain 4 × 0.03 × 0.05 = 0.006 t/h → discoloured share falls from 3% to **0.16%**.
Reject stream: 4 × 0.03 × 0.95 = 0.114 t/h discoloured + 4 × 0.97 × 0.015 = 0.058 t/h good = 0.172 t/h, of which **34% is good grain**.
Passing the reject stream through the sorter again can recover much of that good grain — the same precision–recall trade-off as in Lectures 24 and 25.

## 5. AI across food processing

| Stage | AI application |
|---|---|
| Receiving raw material | Grading lots by size, colour and defects; checking moisture and composition with near-infrared sensing |
| Cleaning and sorting | Detecting and removing foreign objects and impurities (Zhu et al., 2021); colour sorting of grains and seeds (Pasikatan & Dowell, 2001) |
| Processing | Monitoring colour during roasting, frying or drying and adjusting the process; detecting faulty products on the line |
| Packaging | Checking fill level, seals and labels; counting and weighing |
| Storage and distribution | Forecasting demand and shelf life; temperature monitoring in the cold chain (Lecture 23) |
| Whole plant | Predictive maintenance of machines; analysing production data to improve yield and reduce waste |

Computer vision and AI, combined with big data, can give real-time smart machines and predictive models for the food industry and support sustainable food production (Kakani et al., 2020). The food

industry generates vast and complex data, but its ability to analyse and use that data is still limited; AI and machine-learning approaches are being explored to address key challenges (Siddique et al., 2025).

## 6. Benefits and challenges

| Benefits | Challenges |
|---|---|
| Objective, consistent grading at high speed, with records for every batch (Cubero et al., 2011) | High cost of sorting lines and colour sorters for small pack-houses and mills |
| Detection of defects invisible to the eye, using NIR and hyperspectral imaging | Natural variation among varieties and seasons; models need retraining |
| Less manual labour and fatigue; fewer grading disputes | Errors at grade boundaries and missed defects (Figure 3) |
| Removal of contaminated and foreign material improves food safety (Zhu et al., 2021) | Good product lost in reject streams unless re-sorted (Worked example 3) |
| Uniform lots earn better prices and meet AGMARK and export standards (DMI, n.d.) | Need for trained operators, calibration and maintenance |

## 7. Key takeaways

- Sorting and grading raise prices, cut losses, supply processors and protect food safety; AGMARK certifies grades in India under the 1937 Act.
- A grading line singulates, images, analyses, decides and ejects each item; timing links the camera to the ejector.
- Vision systems measure size, shape, colour and defects; NIR and hyperspectral imaging reveal what the eye cannot see.
- Mass depends on the cube of diameter, so small size errors become larger mass errors.
- A multi-class confusion matrix shows grading accuracy and where errors occur — mostly between neighbouring grades.
- Optical colour sorters reject discoloured or infested grains with air jets; their accept and reject streams follow a mass balance.
- AI supports food processing from raw-material grading to packaging, cold chain and maintenance.

## 8. Examination questions

### Short-answer questions (2 marks)

**Q1. Give two reasons for grading agricultural produce.**

Any two: better prices, standard trade descriptions, loss reduction, raw material for processing, food safety.

**Q2. What is AGMARK?**

India's quality certification mark under the Agricultural Produce (Grading and Marking) Act, 1937, acting as a third-party guarantee of quality.

**Q3. Name the five stages of an in-line grading line.**

Feeding and singulation, imaging, analysis, decision and ejection.

**Q4. Name four features measured by a vision grading system.**

Size, shape, colour and external defects (also internal quality with NIR).

**Q5. Why can near-infrared imaging find defects people cannot see?**

It measures reflectance outside the visible spectrum, revealing internal or subtle features.

**Q6. A tomato is 62 mm across; estimate its mass as a sphere of density 1.0 g/cm³.**

(π/6) × 6.2³ ≈ 125 g.

**Q7. How many fruits per hour does one lane carry at 1.0 m/s with cups 0.1 m apart?**

10 per second = 36,000 per hour.

**Q8. What is a multi-class confusion matrix?**

A table of true classes against predicted classes; the diagonal holds correct predictions.

**Q9. How does an optical colour sorter remove bad grains?**

Cameras detect off-colour grains in a falling stream and an air jet blows them into the reject outlet.

**Q10. Name two AI applications in food processing other than sorting.**

Any two: foreign-object detection, process colour monitoring, packaging inspection, demand or shelf-life forecasting, predictive maintenance.

## Long-answer questions (5–10 marks) — answer outlines

**Q1. Explain the need for grading and the role of AGMARK in India.**

Reasons list; AGMARK, 1937 Act, DMI, export and domestic trade, e-NAM laboratories.

**Q2. Describe a machine-vision grading line and the features it measures.**

Figure 1 and stage table; Figure 2; Cubero et al. findings; CNNs.

**Q3. Explain with calculations how size and capacity are determined in a vision grader.**

Worked examples 1 and 2; cube law of mass error; ejector timing.

**Q4. Evaluate a fruit grading system using a confusion matrix.**

Program; Figure 3; accuracy, neighbouring-grade errors, missed defects, false rejects, effect of measurement error.

**Q5. Explain the working of an optical colour sorter for grain with a mass balance.**

Figure 4; visible and NIR detection (Pasikatan & Dowell, 2001); worked example 3; re-sorting.

**Q6. Discuss applications, benefits and challenges of AI in food processing.**

Stage table; Kakani et al., Siddique et al.; benefits and challenges table.

## 9. Glossary

| Term | Meaning |
|---|---|
| AGMARK | India's quality certification mark for agricultural produce |
| Colour (optical) sorter | Machine that ejects off-colour grains with air jets |
| Ejector | Device that pushes an item into its outlet |
| Foreign matter | Material that should not be in food: stones, glass, plastic |
| Grade | Quality class defined by standards |
| Hyperspectral imaging | Imaging in many narrow wavebands |
| In-line inspection | Inspecting every item as it moves along a line |
| Mass balance | Accounting of material in feed, accept and reject streams |
| Multi-class confusion matrix | True classes against predicted classes for more than two classes |
| Near-infrared (NIR) | Light just beyond red, used to sense internal quality |
| Pixel size | Real distance covered by one image pixel (mm/pixel) |
| Reject stream | Material removed by a sorter |
| Singulation | Separating produce into single items for inspection |
| Sorting | Separating items into groups by a property |
| Throughput | Quantity processed per unit time (t/h) |
| Vision grading | Grading by cameras and image analysis |

## References


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Pasikatan, M. C., & Dowell, F. E. (2001). Sorting systems based on optical methods for detecting and removing seeds infested internally by insects or fungi: A review. *Applied Spectroscopy Reviews, 36*(4), 399–416. https://doi.org/10.1081/ASR-100107719

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Zhu, L., Spachos, P., Pensini, E., & Plataniotis, K. N. (2021). Deep learning and machine vision for food processing: A survey. *Current Research in Food Science, 4*, 233–249. https://doi.org/10.1016/j.crfs.2021.03.009


*All diagrams, charts, examples and programs are original. Grade limits, machine values and the simulated batch are illustrative and are not AGMARK standards; every calculation and the program output were checked by running them.*

## Practice MCQs

**Q1. AGMARK certification in India is backed by the**

(a) Food Safety Act 2006
(b) Agricultural Produce (Grading and Marking) Act, 1937
(c) Seeds Act 1966
(d) Essential Commodities Act 1955

**Q2. Separating produce into single items so that each can be inspected is called**

(a) singulation
(b) threshing
(c) blending
(d) curing

**Q3. Which part of a grading line pushes an item into its outlet?**

(a) Camera
(b) Ejector
(c) Hopper
(d) Roller

**Q4. Defects invisible to the human eye can be detected by imaging in the**

(a) audible range
(b) near-infrared and ultraviolet
(c) microwave oven
(d) radio band only

**Q5. At 0.2 mm per pixel, a fruit 310 pixels wide measures**

(a) 31 mm
(b) 62 mm
(c) 155 mm
(d) 0.62 mm

**Q6. A 1.6% error in measured diameter causes an error in estimated mass of about**

(a) 0.5%
(b) 1.6%
(c) 5%
(d) 16%

**Q7. A 4-lane grader with 10 fruits per second per lane handles per hour**

(a) 3,600
(b) 36,000
(c) 144,000
(d) 1,440

**Q8. In a multi-class confusion matrix, correct predictions lie**

(a) in the first row
(b) on the diagonal
(c) in the last column
(d) outside the table

**Q9. In the tomato grading example, most size-grading errors occurred**

(a) between neighbouring grades
(b) between Large and Process
(c) only in Reject
(d) never

**Q10. Optical colour sorters remove bad grains using**

(a) magnets
(b) air jets
(c) water baths
(d) sieves only

**Q11. Internal insect infestation of seeds is generally detectable in the**

(a) visible range
(b) near-infrared range
(c) audible range
(d) X-ray only

**Q12. A sorter fed 4 t/h with 3% bad grain rejects 95% of bad and 1.5% of good grain. The bad-grain share in the accepted rice is about**

(a) 3%
(b) 1.5%
(c) 0.16%
(d) 0%

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**Answers**

Q1 — (b) Agricultural Produce (Grading and Marking) Act, 1937

Q2 — (a) singulation

Q3 — (b) Ejector

Q4 — (b) near-infrared and ultraviolet
Q5 — (b) 62 mm
Q6 — (c) 5%
Q7 — (c) 144,000
Q8 — (b) on the diagonal
Q9 — (a) between neighbouring grades
Q10 — (b) air jets
Q11 — (b) near-infrared range
Q12 — (c) 0.16%